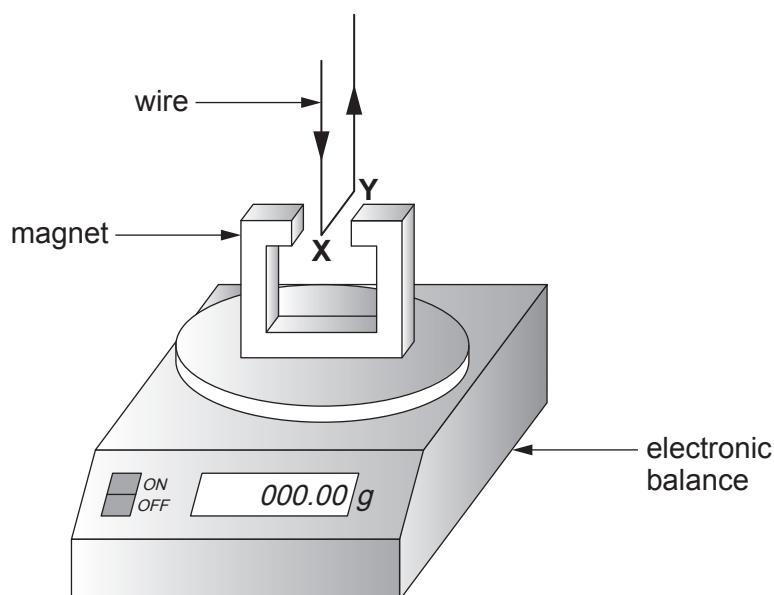


4. The following set-up is used to investigate the force on a current-carrying wire in a magnetic field.



The part **XY** of the wire which is in the magnetic field, has been placed carefully so that the current is from front to back at 90° to the direction of the magnetic field. When there is no current the balance is reset to display 000.00 g. When there is a current in the direction shown, the magnetic force results in a positive reading on the display of the balance. However, this signifies an upward force on the wire.

- (a) (i) Explain briefly why the magnetic force must be upward on the part **XY** of the wire. [2]

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- (ii) **On the diagram**, indicate which is the North pole of the magnet, **and** state below which rule you used to obtain your answer. [2]

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- (iii) Explain why the vertical parts of the wire have no effect on the reading displayed by the electronic balance. [1]

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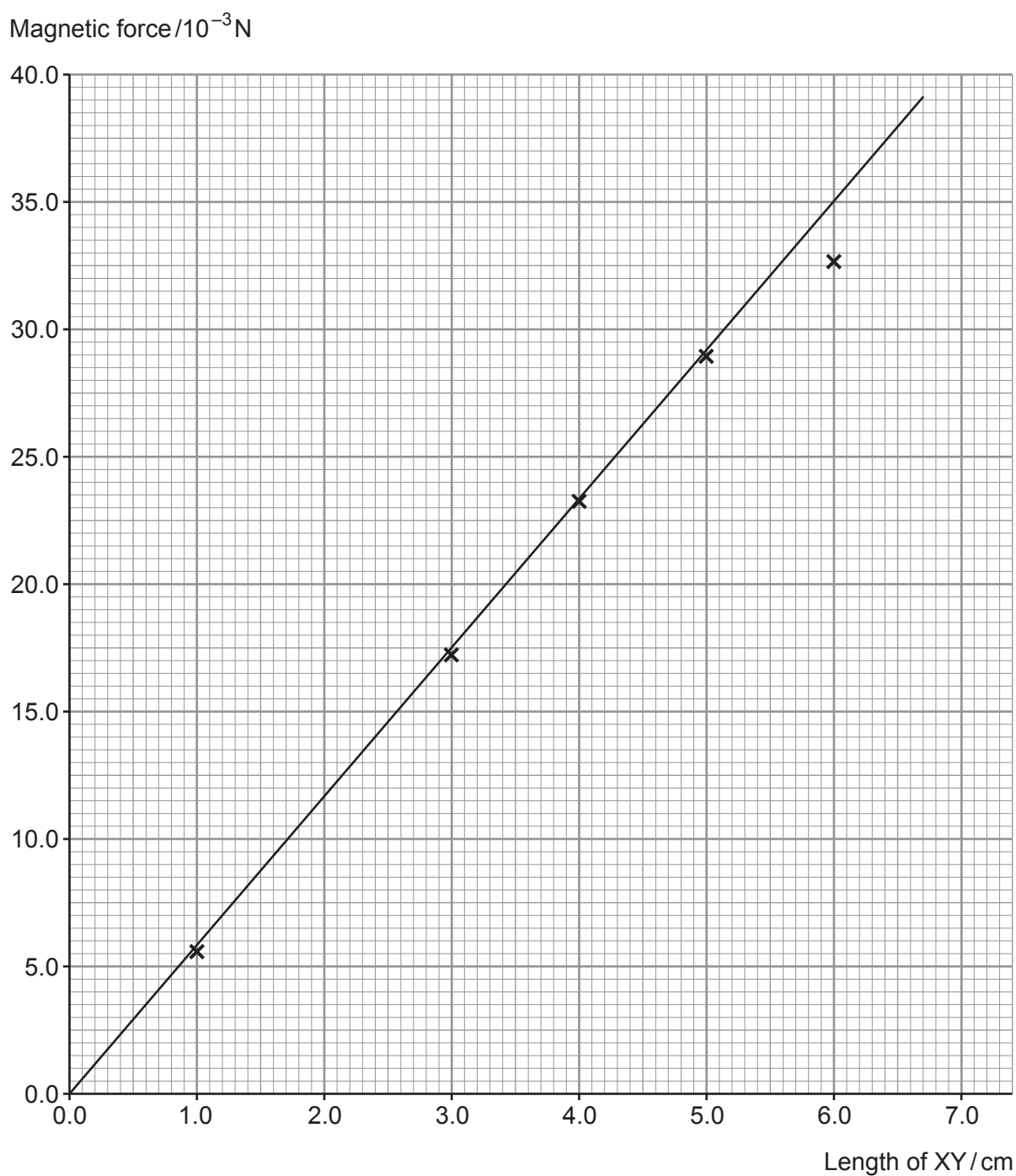


- (b) Bronwen varies the length of XY and records the electronic balance reading each time when a current of 5.00 A flows. She records all her results in a table and plots a graph of magnetic force against length of wire.

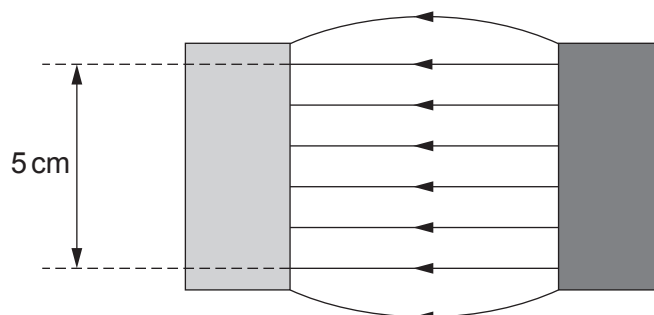
- (i) **Complete the table and plot** the two missing points on the graph. [4]

Length of XY/cm	Balance reading/g	Magnetic force/ 10^{-3} N
1.0	0.56	5.5
2.0	1.22	
3.0	1.75	17.2
4.0	2.36	23.1
5.0	2.89	28.4
6.0	3.32	32.6
7.0		35.8





- (ii) The region of uniform magnetic field between the poles of the magnet is of length 5 cm only. Evaluate whether or not Bronwen has drawn the line of best fit correctly. [3]



- (iii) Determine the magnetic flux density, B , of the uniform magnetic field between the poles of the magnet and quote it to an appropriate number of significant figures. The current in the wire is 5.00 A. No uncertainty is required in your value of B . [5]

